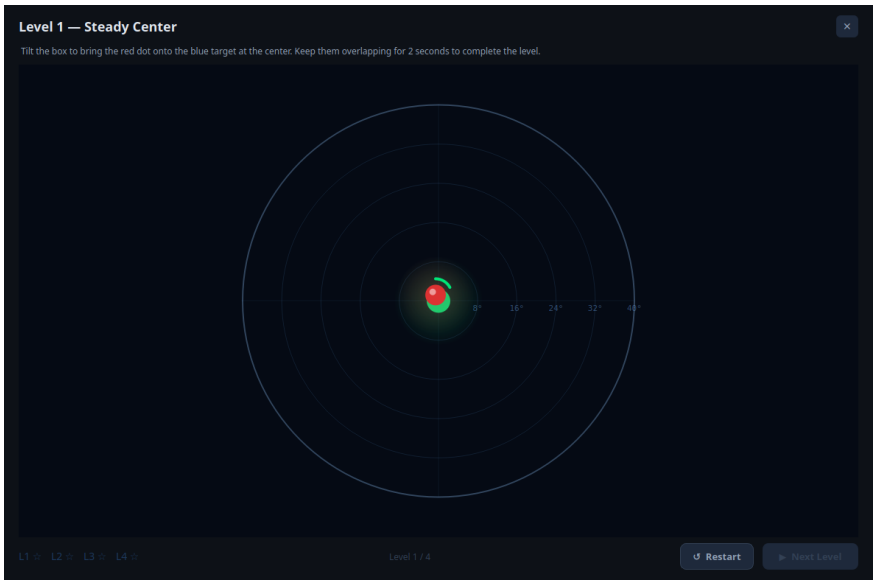
Gyro Target

Four levels, the last two with the controls reversed — and a stopwatch that measures the brain learning

Level: Explorer and Scientist — ages 11–19 (Romanian school years V–XII) • **Time in class:** 20 min as a hook, or a 50-minute lesson • **Sensors:** MPU-6500 (acceleration)



The activity screen. The values shown are simulated; yours will differ.

Before you start

Who it is for	Explorer and Scientist — ages 11–19. As a game it works at any age. The **content**, though, rests on the nervous system and reflexes — around **age 13** (year VII) — while the physics side, gravity as a reference direction and the components of a vector with sine and cosine, becomes accessible only after year VII, when the trigonometric functions arrive.
Assumed knowledge	Nothing at all is needed to play. For the full discussion: the idea that an accelerometer senses gravity (the *Among the Clouds* activity is the natural preparation) and the notion of a reflex.
Time in class	Short: 20 min, the first two levels and the shock of level three. Full: a 50-minute lesson, with **the learning curve measured** on level three — the most valuable part of the activity.
Teacher preparation	5 min — play it yourself as far as level four, **so that you know how disorienting level three is**. A teacher who has not been through that feeling cannot announce it credibly. Have a stopwatch ready for the learning-curve version.
Sensor settling time	**None.** The accelerometer responds at once. The red dot is **smoothed by about a tenth of a second**, though: it moves slightly behind the box, and that lag is exactly why sharp corrections end in oscillation.
Sensors and boxes	MPU-6500. One box is enough, with the screen projected and the children taking turns. **The box is tilted, not carried through the air.**
Three routes	

Short	20 min: the prediction, levels 1 and 2, then level 3 with the controls reversed, and the debrief about what happened in the brain.
Full	The 50-minute lesson: the short route, plus **four timed attempts at level 3**, the graph of time against attempt, and level 4 to finish.
Project	Extension: the flat-rotation test — *can the box tell that I am turning it on the table?* — which shows the sensor's real limit.

1. Where this fits the curriculum

- **Biology** — around **age 13**, the nervous system, reflexes and the coordination of movement. The reversed-controls level **makes visible** something that otherwise stays a definition: a learned reflex does not stop on command — it has to be *relearned*.
- **Physics** — in the upper school, gravity as a reference direction and the resolution of a vector into components: the box works out its tilt from the ratio of the acceleration components.
- **Mathematics** — angles and coordinates (the arena's rings are marked in degrees); the sine and cosine of the science background belong to year VII and above; and on the full route, **a graph of time against attempt number**, which is a graph with a real meaning rather than an exercise.
- **Physical education** — hand-eye coordination and fine motor control; the game demands both and also measures them.
- **Across subjects** — the idea of a *feedback loop with a delay*: you look, you correct, but the correction arrives a fraction of a second late. It is the same problem at a steering wheel, in a robot and in a thermostat.

The central skill is **watching your own learning**. The pupil does not merely discover that the reversed level is hard; they measure it four times and *see* the time fall. Few activities put a stopwatch on a change inside the pupil's own brain.

2. Timing

Route	Time	What happens
Hook	**20 min**	The prediction (3), level 1 for the control map (4), level 2 (4), level 3 and the class's reaction (6), the debrief (3).
Full	**50 min**	The short route (20), **four timed attempts at level 3** and their graph (15), the flat-rotation test (5), level 4 and the conclusion (10).

Level 3 is the activity; the first two are the preparation

Levels 1 and 2 look like the game, but they are really **the training that makes level 3 hurt**. They install a map of the controls within a few minutes, and level 3 overturns it — which is when you see how solid something learned only two minutes earlier already was. That is why **you must not skip the first two levels**, even when time is short and they look easy. A pupil who goes straight to the reversed controls has nothing to lose and feels nothing.

And **do not announce the reversal in advance**. Let them discover it in the first seconds of level 3: that burst of laughter in the room is exactly the reaction the activity is built on, and the debrief starts from it.

3. Materials

What you need	Per group	Notes
MagicBox with MPU-6500	1	It is **tilted**, not carried through the air.
Laptop with the app	1	**WiFi is preferable**: the box turns a great deal in the hands, and the cable twists.
Stopwatch or phone	1 per group (recommended)	**Not strictly required**: the app keeps *every* attempt, but lists them only in the results window at the end, and **Play Again** clears them. The stopwatch puts each time on the sheet as it happens, it is an independent check — and a pupil who measures for themselves remembers differently from one who reads a screen.
Pupil sheet, printed	1 per pupil	The table of four attempts is there.
Neck strap or wrist strap	1	The box turns in every direction and slips easily.

Preparation. Play through to level four. Not to get good at it, but to go through the disorientation of level three — it is the only way you can tell the class, honestly, *I know exactly how that feels.*

4. Science background for the teacher

The name says gyroscope, but the box uses gravity

The app is called *Gyro Target*, and the MPU-6500 really does contain a gyroscope — **but this game does not use it**. It reads the three components of the acceleration, a_x, a_y and a_z, that is **the direction in which gravity pulls** relative to the box, and works out two angles from them:

$$\text{roll} = \arctan\left(\frac{a_y}{a_z}\right), \qquad \text{pitch} = \arctan\left(\frac{-a_x}{\sqrt{a_y^2 + a_z^2}}\right).$$

A box lying flat feels all of gravity on its vertical axis; tilted, it shares gravity between axes, and the ratio of the components gives the angle.

The consequence can be tested in class, and is worth testing: if you turn the box *flat on the table*, gravity pulls the same way in every one of those positions — so **the red dot does not move**. An accelerometer can see tilt, but **it cannot see rotation about the vertical**. That is exactly the gyroscope's job, and this game, despite its name, never asks for it.

The largest useful tilt is 40°: at that angle the red dot reaches the outer ring, the one marked 40°, and beyond it goes no further. The formulas are for the teacher; with lower-secondary pupils the idea is put into words, and the calculation with the tangent is kept for those who have met trigonometry, from year VII.

4.1 Why sharp corrections get out of hand

The red dot does not follow the box instantly: its position is smoothed, with a lag of **about a tenth of a second**. That is little — and yet enough to turn the game into a classic control problem.

When a pupil sees the dot near the target and corrects sharply, the correction takes effect *after* the dot has already passed the target. The pupil corrects again, harder, the other way, and **oscillation** sets in: the dot crosses from one side to the other, further each time, never staying still for the two seconds required.

The strategy that works is counter-intuitive and worth stating out loud after the first burst of frustration: *small movements, and pauses.* Tilt a little, **wait** to see the effect, then correct. It is precisely what a good driver does on ice and a good crane operator does at the controls — a real skill, not a trick of the game.

4.2 What happens at the reversed levels

At levels 3 and 4 both axes are flipped: *tilt up and the dot goes down; tilt left and it goes right.* The difficulty does not come from the rule being hard to grasp — anyone grasps it in three seconds — but from the fact that **the hand does not obey the understanding**. Correcting an error seen on screen happens by reflex, faster than thought, and the reflex uses the old map.

What follows, if the pupil persists, is **adaptation**: after a few attempts the time falls visibly. That is the heart of the activity, and it can be *measured.*

What the screen says about adapting — and what can be checked

At level 3 the app displays a message saying that the "up means up" reflex was built over years and does not switch off because you know the rule — and telling the pupil to watch their own times, because they usually fall over the next few tries.
That message replaced one that put a number on it: it used to say that rewriting the map takes *three to seven days of daily practice.* That was not a measured constant, and children read numbers on screens as facts. It is worth telling a class why it changed — a good lesson in how to read a figure an instrument prints.
What can honestly be claimed is more interesting anyway: adaptation to a reversed map **begins within minutes** — visible in the lesson itself, in the falling times, which the app now shows by itself — but **it does not end there**; going back to the normal mapping produces a brief clumsiness of its own, a sign that the old map has not vanished but has been competed with.
The right way to put it to a class: *your brain is starting to adapt right now, during this lesson — and that is why the second attempt is faster than the first. How long a complete change takes depends on the person and on how much they practise.*

5. Running the lesson

1. **The prediction (3 min).** Before playing: *how do you think the red dot is controlled? And if I tilt the box to the left, where do you think it will go?* The prediction is written down and not changed afterwards.
2. **Level 1 — the control map (4 min).** The first thing to do is not to win but to **establish the map**: tilt one direction at a time and note where the dot goes. The sheet has a table for exactly this.

Technical note — the four levels, the times and the sensor's limit

The sensor. MPU-6500, but **the accelerometer only**: the game reads a_x, a_y, a_z and infers tilt from the direction of gravity. The box is **tilted**; it is not carried through the air, because a sharp movement adds an acceleration of its own and the dot jumps.
How to start. From the **Activities** page, on the **Gyro Target** card, with the **Start** button. You choose a box when it opens. From firmware 2.2.5 the chip smooths its own

readings at about 20 Hz, so the dot trembles less.
The four levels. Level 1: Steady Center; **Level 2: Moving Target**; **Level 3: Inverted Controls**; **Level 4: Inverted & Moving**. In each, the red dot must be **held over the target for two unbroken seconds**; if it slips off, the overlap timer restarts from zero.
How precise it has to be. The tolerance is of the order of **four or five degrees** of tilt. And the largest useful tilt is 40°, exactly the outer ring: beyond that the dot goes no further.
The arena's rings. The five rings are marked every eight degrees — 8°, 16°, 24°, 32° and 40° on the outer ring —, and their labels now read clearly on a projector too. With them the control map can also be recorded in numbers: *I tilted about* 16° *and the dot reached the second ring.*
A quirk of the window. The tolerance also has a fixed part, in pixels, so **a small window is slightly more forgiving** than one stretched across the whole screen. If a group is struggling needlessly, make the window smaller — it is not cheating, it is the same task at a different scale.
The moving target traces a slow curve and **stays within the inner half of the arena**: it never reaches the outer ring, so a moderate tilt is always enough. The curve does not repeat itself within a lesson.
The times and the score. The app keeps **every attempt**, not just the best one, and the results window shows the run in the order it happened — for example 41.2 → 28.7 → 19.4 *s*. The star beside the level number still shows the best time. **The run is the data that matters**, because the trend is in it.
The score falls continuously with time: half marks at twenty seconds, and less after. **It no longer stops at a floor**, as it used to — it stopped at 100 points after 45 seconds, so every slower attempt tied, precisely among the pupils finding it hardest. Even so, **the times say more than the points**, because they compare directly between attempts.
The buttons. Restart keeps the best time; **Next Level** becomes available once the level is complete; at the end, **See Results** opens a window with the times, the scores and the totals, offering **Play Again**, which *clears every time*, and **Close**. There is no export — which is why the learning curve is written on the sheet, with a stopwatch or copied from the results window before anyone presses **Play Again**.
The sensor's limit, to test in class. Held flat and turned on the table, the box **does not move the dot**: gravity pulls the same way in all those positions.

3. **Level 2 (4 min).** The target moves. This is where the good strategy first appears: *follow it with small movements, do not chase it.*
4. **Level 3 — the reversal (6 min).** With no warning. Let the first thirty seconds pass without intervening; then name what is happening and move on to measuring.
5. **The four timed attempts (full route, 15 min).** Each pupil plays level 3 four times in a row, with a classmate on the stopwatch, and records the four times. **This is the part most worth the class time.**
6. **The flat-rotation test (full route, 5 min).** The box flat on the table, turned slowly. The dot does not move — and why it does not is the whole physics of the activity.
7. **Level 4 (full route).** As a finish, not as a goal. Not everyone completes it, and they do not need to.

Debrief: the score is not the point

The first question is not about the score. It is: **"what did your hand do in the first seconds of level 3?"**

The answers are the content of the lesson: *I moved exactly the wrong way*; *I knew the rule, but my hand didn't*; *I had to stop and think*. The last is the most valuable: replacing a reflex with a conscious decision is precisely what happened.

The four questions, in order:

- *Did you understand the level 3 rule straight away?* — yes, everyone did. *And yet your hand kept getting it wrong. Why?* — because the correcting movement is made by reflex, faster than thought, and the reflex was using the old map.
- *Was the fourth attempt faster than the first?* — almost always. **That is the brain adapting, measured with a stopwatch.**
- *What strategy actually worked?* — small movements and pauses. The dot answers with a slight delay, and sharp corrections produce oscillation.
- *How does the box know how you are holding it?* — from the direction in which gravity pulls. And if you turn it flat on the table it knows nothing at all — the test you did.

One closing question, if there is time: *where else does a person have to fight a learned reflex?* Their answers — the rear-view mirror, driving on the other side of the road, a changed keyboard — are exactly the same situation.

And a note on fairness. Times differ widely between pupils, for reasons that have nothing to do with effort. **Compare each pupil with themselves**, first attempt against fourth, never with their neighbour.

The investigation, in one table

The question. Does a person get better at reversed controls within a few minutes — or stay just as clumsy?

The hypothesis. Written **before** the fourth attempt: on the fourth try I will be _____ than on the first. It is not changed afterwards.

What I change. Nothing, deliberately: **the only variable is practice**. The same level, the same box, four attempts one after another.

What I measure. The time from the start of the attempt to success, on a classmate's stopwatch.

What I keep the same. The same body position, the same distance from the screen, the same hand, and **short, equal pauses** between attempts. If you swap hands at the third attempt, you are measuring something else.

How many repeats. Four attempts, so that a trend shows rather than just a difference. With two points you cannot tell learning from luck.

What makes an observation sound — and what makes it an artefact. Sound is a run of four consecutive attempts, under the same conditions, with the times written down as they happen. An artefact is: an attempt in which the box was *carried* rather than tilted, because then the dot jumps for another reason; a long pause in the middle of the run; an unannounced change of strategy — switching from sharp movements to small ones, say, which shortens the time by itself and has nothing to do with adaptation; and **a time that *rises* on the last attempt through tiredness or laughter**, which does not disprove adaptation but shows that we are measuring a person, not an instrument.

6. After the activity

6.1 What the observations should look like

What I do	What happens	What it means
Level 1, big tilts	The dot overshoots both ways	Oscillation: the corrections are larger than the error.
Level 1, small tilts with pauses	The dot settles and stays	The right strategy, and the hard one to accept.
Level 2	It works, but needs continuous tracking	The target stays in the inner half; large tilts are never needed.
Level 3, first seconds	The hand does exactly the opposite	The reflex is faster than the understood rule.
Level 3, fourth attempt	A visibly shorter time	**Adaptation, measured within one lesson.**
Back to level 1 after level 3	A moment of clumsiness	The old map was competed with, not erased.
Box flat, turned on the table	**The dot does not move**	Gravity cannot see rotation about the vertical.
Box moved sharply through the air	The dot jumps	The acceleration of the movement adds to gravity.

6.2 The learning curve, step by step

1. **Work in pairs.** One plays, the other times and records. The roles swap after the four attempts.
2. **Start the stopwatch with the level** and stop it when the success message appears. **Do not use the star** — it reports only the best attempt, which is precisely the figure that hides the curve. The full run comes either from the stopwatch or from the app's results window; **best of all from both**, compared.
3. **Four attempts in a row**, with short, equal pauses.
4. **Write down all four times**, the poor ones included. A run with the failed attempt deleted shows nothing at all.
5. **Draw the graph**: attempt number across, time up. **The downward trend is the result**, not any single value.
6. **Compare trends between classmates, not times.** Someone who went from 40 seconds to 20 learned as much as someone who went from 20 to 10.

6.3 Questions to ask afterwards

1. You knew the level 3 rule from the first second. Why was that not enough?
2. Compare your first attempt with your fourth. What changed — the rule, the hand, or the attention?
3. Why do sharp corrections make the dot overshoot?

4. You turned the box flat on the table and the dot did not move. What does that tell us about what the sensor actually measures?
5. Give an example from your own life where a learned reflex gets in your way when the rules change.

6.4 Assessment ideas as you go

- **Low stakes:** the pupil fills in the level 1 control map correctly — what the dot does for each direction of tilt.
- **Standard:** the pupil records the four times, draws the graph and says which way the trend goes and why.
- **Extension:** the pupil explains why small movements work better than large ones, using the idea of a delay, and interprets the flat-rotation test correctly.

Common pitfalls and misconceptions

Carrying the box through the air. An accelerometer cannot tell gravity from the acceleration of a sharp movement, so the dot jumps for no apparent reason. The box is **tilted**, held roughly in place.
Large corrections. They are the natural reflex and they are wrong: the dot responds with a delay, and too large a correction produces oscillation. Small movements, with pauses.
Announcing the reversal in advance. It spoils half the activity.
Skipping levels 1 and 2 "to get to the interesting part". Without the map installed by the first two levels, level 3 overturns nothing.
Using the score instead of the times. The score now falls continuously, so it no longer ties the slow attempts — but it is still a derived quantity. Times compare directly and can go on a graph; points cannot.
Comparing pupils with each other. The starting differences are large and do not measure effort. Each pupil is compared with themselves.
Clearing the times by accident. Play Again restarts the whole game and erases every time. Write them on the sheet first.
The USB cable. The box turns in every direction; the cable twists and tugs. Use WiFi.

Safety and classroom management

A wrist or neck strap is recommended. The box turns in the hands and slips easily.
The player stays seated, elbows on the table. That position is not only safer but better for the game: the support reduces tremor.
The screen is dark and the dots are small. Seat pupils with visual difficulties close by and enlarge the window; check the screen brightness too.
A dot oscillating against a dark background can disturb pupils who are sensitive to motion or who have a history of seizures triggered by visual stimuli. **Announce the activity in advance**, allow anyone to withdraw without explaining, and offer them the timekeeper's role.
The frustration at level 3 is real. Say at the outset that *nobody does it well the first time* and that *no marks are given for times* — otherwise the activity becomes, for some pupils, an experience of public failure.
Limit the number of attempts at level 4. It is designed to be hard, and persisting beyond a few minutes teaches nobody anything.

Adaptations for pupils with special educational needs

The box supported on the table. The most useful adaptation: the box is held in both hands with the elbows resting. It reduces tremor and fatigue and **improves everyone's result**, not only that of pupils with motor difficulties.
A smaller window. It makes the tolerance a little larger. This is a legitimate adaptation, easy to make and invisible to classmates.
Levels 1 and 2 without 3 and 4. For pupils for whom the frustration of the reversal is not productive, the first two levels are a complete activity in themselves, with all the physics content.
Three real roles, rotated. The **player**; the **timekeeper**, who starts, stops and calls out the time; the **scribe**, who records and draws the graph. The timekeeper has a task of precision, not a decorative one.
Without a hand-held stopwatch. If operating a stopwatch is itself a barrier, copy the times from the app's results window at the end: it lists each level's attempts in order (the last four).
For pupils sensitive to the motion on screen, the scribe's role does not require a fixed gaze on the arena.

If things do not go to plan

The red dot does not move at all. The box is not sending acceleration data: check the connection, and that values are showing for AX, AY and AZ on the **Live Data** page. **If it moves only when you tilt the box and not when you turn it flat on the table, everything is fine** — that is how it should be.
The red dot sits stuck to the edge of the arena while the box lies flat. The box has its motion sensor mounted the other way up: lying on the table it shows about −1 g on AZ instead of +1 g. The game is not at fault. In **Settings → Box names** tick **Motion sensor upside down** for that box; its data are then turned the right way up in every activity. With firmware 2.3.0 or newer the tick is kept in the box itself, so it counts on every computer; the box must be connected when you tick it.
The dot trembles or jumps. The box is being carried rather than tilted, or the connection is poor. Rest the elbows on the table and switch to WiFi.
Nobody gets past level 3. Make the window smaller, move to the box-supported-on-the-table position, and remind them of the strategy: **small movements, with pauses**. Level 3 is solvable by anyone with the right strategy; with sharp corrections, by nobody.
The times were cleared by accident. The learning curve lives on the sheet, not in the app — which is why it is written down as it happens.
One box and thirty pupils. Project the screen, swap pairs, and use the three roles. Level 1 takes under a minute for a practised pupil, so the turn comes round quickly.
The box will not connect. Do not stop the lesson — do the version with no apparatus: each pupil writes on a sheet of paper while watching only *its reflection in a mirror*, holding the paper under their chin, or draws a star seen only in the mirror. **It is exactly the same conflict between reflex and rule**, it is a classic task from psychology laboratories, and it can be timed the same way, over the same four attempts.

7. Further resources

Stable, well-established starting points. Search each site for the subject named — these are top-level addresses, not deep links, so they do not rot.

- **BrainFacts** — `https://www.brainfacts.org` — search for neuroplasticity and motor learning; written for a general audience and backed by the neuroscience society.

- **Nature Education, Scitable** — `https://www.nature.com` — search for synaptic plasticity; useful for the upper school.
- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for the simulations on vectors and the resolution of forces, for the physics of tilt.
- **InvenSense / TDK** — `https://invensense.tdk.com` — the MPU-6500 documentation, if you want to see the difference between an accelerometer and a gyroscope at source.
- **STEM Learning** — `https://www.stem.org.uk` — search for the mirror-drawing experiments on hand-eye coordination; they are the no-apparatus version of this activity.
- **Institute of Physics** — `https://www.iop.org` — search for the material on accelerometers and the sensors inside phones.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *you knew the level 3 rule from the first second — why was that not enough?* Collect it in the last two minutes.

For a more formal judgement, use the **shared three-strand rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. Here the three strands look like this:

- **Measurement practice** — the pupil times consistently, makes the four attempts under the same conditions, and records every time, the poor ones included.
- **Data interpretation** — the pupil draws the time-against-attempt graph and describes the trend, without mistaking one good value for learning.
- **Scientific reasoning** — the pupil explains why a beaten reflex needs practice, why small movements work, and what the flat-rotation test shows about the sensor.

The rubric itself, with the descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. The pupil sheet

The sheet the pupils fill in is a separate document: `docs/en/student_sheets/gyro_target.pdf`. It runs to two pages — the control map and the prediction on the first; the four timed attempts, the graph and the rotation test on the second — so it prints double-sided, one sheet per pupil.

Three things to know before printing it. **Each of the four times has a row of its own**, and beneath them is a small grid for drawing the graph — for many pupils this is the first graph they have made from their own data, so the grid comes ready-made, with the axes named. **The sheet asks explicitly for the poor times too**: a run with the bad attempt deleted shows no trend at all, and that is said on the sheet, in the pupil's own words. **The flat-rotation test has a row of its own**, with the prediction kept separate from the result, because it is the only place where a pupil tests a *limit* of the sensor rather than a capability of it. The tables are not repeated here precisely so that the two documents cannot drift apart: there is one printable version, and it is the sheet.